

JESS Thermodynamic Database v8.9

Chemical species in reactions with Hafnium

24-Jan-23

Charge	CAS	Count	Molecular formula	Mol. mass
234TriOHPentDioic-2				
-2	---	39	C(5)H(6)O(7)	178.098
2ThenTriFAcone-1				
-1	---	101	C(8)H(4)F(3)O(2)S(1)	221.174
Acetic-1 Acetate ion				
-1	71-50-1	558	C(2)H(3)O(2)	59.0446
B(s) Boron; Boron, rhombohedral; Boron, beta				
0	7440-42-8	111	B(1)	10.8100
Ba(s) Barium; Barium, cubic; Barium, lambda trans.				
0	7440-39-3	64	Ba(1)	137.330
BaHfO3(s) Barium hafnium trioxide				
0	---	1	Ba(1)Hf(1)O(3)	363.820
Br-1 Bromide ion				
-1	24959-67-9	1213	Br(1)	79.9040
Br2(l) Bromine, liquid				
0	---	285	Br(2)	159.808
C(s) Carbon; Graphite; Carbon, hexagonal				
0	7440-44-0	530	C(1)	12.0110

Ca (s) Calcium; Calcium, cubic; Calcium, fcc; Calcium, alpha				
0	7440-70-2	184	Ca (1)	40.0780
CaO.HfO2 (s) Calcium hafnium trioxide; 1-Calcium oxide 1-hafnium oxide				
0	---	1	Ca (1) Hf (1) O (3)	266.568
Citric-3 Citrate ion				
-3	126-44-3	1347	C (6) H (5) O (7)	189.102
Cl-1 Chloride ion				
-1	16887-00-6	2242	Cl (1)	35.4530
Cl2 (g) Chlorine gas				
0	7782-50-5	589	Cl (2)	70.9060
DTPA-5 Diethylenetrinitrilotetraacetate ion; 1,4,7-triazaheptane-1,1,7,7-pentaacetate ion; N,N-bis[2-[bis(carboxymethyl)amino]ethyl]glycinate ion; DTPA ion				
-5	---	343	C (14) H (18) N (3) O (10)	388.311
e-1 Electron				
-1	---	3399	E (1)	0.000000
EDTA-4 Ethylenediaminetetraacetate anion				
-4	---	1457	C (10) H (12) N (2) O (8)	288.214
F-1 Fluoride ion				
-1	16984-48-8	1090	F (1)	18.9984
F2 (g) Fluorine gas				
0	7782-41-4	259	F (2)	37.9968

Gly-1 Glycinate ion; Aminoacetate ion				
-1	---	1356	$C(2)H(4)N(1)O(2)$	74.0593
Glycolic-1 Glycolate ion				
-1	57122-18-6	342	$C(2)H(3)O(3)$	75.0440
H+1 Hydrogen ion; Proton				
1	12408-02-5	31679	$H(1)$	1.00794
H+1_Ascorbic-2				
-1	---	53	$C(6)H(7)O(6)$	175.118
H+1_F-1				
0	---	142	$F(1)H(1)$	20.0063
H+1_Gluconic-1 Gluconic acid; D-Gluconic acid; Dextronic acid; Maltonic acid; Glyconic acid; Glycogenic acid; Pentahydroxycaproic acid; D-2,3,4,5,6-Pentahydroxyhexanoic acid				
0	526-95-4	4	$C(6)H(12)O(7)$	196.157
H+1_Gly-1 Glycine; Aminoacetic acid; Aminoethanoic acid; 2-Aminoethanoic acid				
0	---	82	$C(2)H(5)N(1)O(2)$	75.0672
H+1_Mucic-2				
-1	---	4	$C(6)H(9)O(8)$	209.133
H+1_PO4-3 Hydrogenphosphate ion				
-2	---	317	$H(1)O(4)P(1)$	95.9795
H+1_Salicylic-2				
-1	---	111	$C(7)H(5)O(3)$	137.115
H+1_Ser-1 Serine; L-Serine; beta-Hydroxyalanine; L-3-Hydroxy-2-aminopropionic acid; 2-Amino-3-hydroxypropanoic acid				

0	---	6	C(3)H(7)N(1)O(3)	105.094
H+1_SO4-2 Hydrogensulfate ion				
-1	---	83	H(1)O(4)S(1)	97.0655
H+1_SulfSal-3				
-2	---	54	C(7)H(4)O(6)S(1)	216.165
H+1_TriMDTA-4				
-3	---	20	C(11)H(15)N(2)O(8)	303.249
H+1(2)_234TriOHPentDioic-2 2,3,4-Trihydroxypentanedioic acid; Xylotrihydroxyglutaric acid; Trihydroxyglutaric acid				
0	---	5	C(5)H(8)O(7)	180.114
H+1(2)_Citric-3				
-1	---	39	C(6)H(7)O(7)	191.117
H+1(2)_Lactic-1				
1	---	5	C(3)H(7)O(3)	91.0868
H+1(2)_Malic-2 Malic acid; Hydroxybutanedioic acid; L-Malic acid; L-Hydroxybutandioic acid; (S)-(-)-Hydroxysuccinic acid; (S)-(-)-Malic acid; 2-Hydroxysuccinic acid; Hydroxyethane-1,2-dicarboxylic acid				
0	97-67-6	18	C(4)H(6)O(5)	134.089
H+1(2)_Oxalic-2				
0	144-62-7	33	C(2)H(2)O(4)	90.0355
H+1(2)_Tartaric-2 Tartaric acid; L-Tartaric acid; d-Tartaric acid (natural); dextroTartaric acid; 2,3-Dihydroxybutanedioic acid; Dihydroxysuccinic acid				
0	87-69-4	22	C(4)H(6)O(6)	150.088
H+1(2)_Tiron-4				
-2	---	27	C(6)H(4)O(8)S(2)	268.213

H+1 (2) _TTHA-6				
-4	---	11	C (18) H (26) N (4) O (12)	490.424
H+1 (3) _NTA-3 Nitritotriacetic acid; N,N-Bis(carboxymethyl)glycine; Triglycollamic acid; Trimethylaminetricarboxylic acid; Tri(carboxymethyl)amine; Triglycine; NTA; Nitritotriethanoic acid				
0	139-13-9	10	C (6) H (9) N (1) O (6)	191.141
H+1 (3) _Pyrogallol-3 1,2,3-Trihydroxybenzoic acid; 1,2,3-Benzenetriolic acid; Pyrogallol; 1,2,3-Benzenetriol				
0	87-66-1	9	C (6) H (6) O (3)	126.112
H2O Water				
0	7732-18-5	5947	H (2) O (1)	18.0153
H2O2 Hydrogen peroxide, aqueous				
0	---	27	H (2) O (2)	34.0147
Hf+4 Hafnium(IV) ion				
4	22541-25-9	148	Hf (1)	178.492
Hf+4 _2ThenTriFAcne-1				
3	---	1	C (8) H (4) F (3) Hf (1) O (2) S (1)	399.666
Hf+4 _2ThenTriFAcne-1 (2)				
2	---	1	C (16) H (8) F (6) Hf (1) O (4) S (2)	620.840
Hf+4 _2ThenTriFAcne-1 (3)				
1	---	1	C (24) H (12) F (9) Hf (1) O (6) S (3)	842.013
Hf+4 _2ThenTriFAcne-1 (4)				
0	---	1	C (32) H (16) F (12) Hf (1) O (8) S (4)	1063.19
Hf+4 _Acetic-1_OH-1 (2)				

1	---	2	C (2) H (5) Hf (1) O (4)	271.551
Hf+4_Acetic-1_OH-1 (3)				
0	---	2	C (2) H (6) Hf (1) O (5)	288.559
Hf+4_Acetic-1 (2)_OH-1 (3)				
-1	---	2	C (4) H (9) Hf (1) O (7)	347.603
Hf+4_Br-1				
3	---	1	Br (1) Hf (1)	258.396
Hf+4_Br-1 (4)_ (s) Hafnium tetrabromide				
0	---	1	Br (4) Hf (1)	498.108
Hf+4_Citric-3				
1	---	1	C (6) H (5) Hf (1) O (7)	367.594
Hf+4_Citric-3 (2)				
-2	---	1	C (12) H (10) Hf (1) O (14)	556.695
Hf+4_Cl-1				
3	---	2	Cl (1) Hf (1)	213.945
Hf+4_Cl-1_SO4-2				
1	---	1	Cl (1) Hf (1) O (4) S (1)	310.003
Hf+4_Cl-1 (2)				
2	---	3	Cl (2) Hf (1)	249.398
Hf+4_Cl-1 (3)				
1	---	3	Cl (3) Hf (1)	284.851
Hf+4_Cl-1 (4)				
0	---	2	Cl (4) Hf (1)	320.304
Hf+4_Cl-1 (4)_ (g) Hafnium tetrachloride gas				

0	---	1	Cl (4) Hf (1)	320.304
Hf+4_Cl-1 (4)_(s) Hafnium tetrachloride				
0	---	2	Cl (4) Hf (1)	320.304
Hf+4_DTPA-5				
-1	---	1	C (14) H (18) Hf (1) N (3) O (10)	566.803
Hf+4_EDTA-4				
0	---	3	C (10) H (12) Hf (1) N (2) O (8)	466.706
Hf+4_F-1				
3	---	3	F (1) Hf (1)	197.490
Hf+4_F-1 (2)				
2	---	4	F (2) Hf (1)	216.489
Hf+4_F-1 (3)				
1	---	4	F (3) Hf (1)	235.487
Hf+4_F-1 (4)				
0	---	5	F (4) Hf (1)	254.486
Hf+4_F-1 (4)_(s) Hafnium tetrafluoride; Hafnium tetrafluoride, monoclinic				
0	---	2	F (4) Hf (1)	254.486
Hf+4_F-1 (5)				
-1	---	5	F (5) Hf (1)	273.484
Hf+4_F-1 (6)				
-2	---	4	F (6) Hf (1)	292.482
Hf+4_Gluconic-1				
3	---	1	C (6) H (11) Hf (1) O (7)	373.641
Hf+4_Glycolic-1				

3	---	2	C(2)H(3)Hf(1)O(3)	253.536
Hf+4_Glycolic-1_OH-1(2)				
1	---	1	C(2)H(5)Hf(1)O(5)	287.551
Hf+4_Glycolic-1(2)				
2	---	1	C(4)H(6)Hf(1)O(6)	328.580
Hf+4_H+1_234TriOHPentDioic-2				
3	---	1	C(5)H(7)Hf(1)O(7)	357.598
Hf+4_H+1_Ascorbic-2				
3	---	1	C(6)H(7)Hf(1)O(6)	353.610
Hf+4_H+1_Citric-3				
2	---	1	C(6)H(6)Hf(1)O(7)	368.601
Hf+4_H+1_Gly-1				
4	---	2	C(2)H(5)Hf(1)N(1)O(2)	253.559
Hf+4_H+1_Lactic-1				
4	---	1	C(3)H(6)Hf(1)O(3)	268.571
Hf+4_H+1_Malic-2				
3	---	1	C(4)H(5)Hf(1)O(5)	311.573
Hf+4_H+1_OH-1(2)_Pyrogallol-3				
0	---	1	C(6)H(6)Hf(1)O(5)	336.603
Hf+4_H+1_OH-1(3)_Salicylic-2				
0	---	1	C(7)H(8)Hf(1)O(6)	366.629
Hf+4_H+1_OH-1(3)_Ser-1				
1	---	1	C(3)H(10)Hf(1)N(1)O(6)	334.608
Hf+4_H+1_OH-1(3)_SulfSal-3				
-1	---	1	C(7)H(7)Hf(1)O(9)S(1)	445.679

Hf+4_H+1_Oxalic-2				
3	---	1	C(2)H(1)Hf(1)O(4)	267.520
Hf+4_H+1_Tartaric-2				
3	---	1	C(4)H(5)Hf(1)O(6)	327.572
Hf+4_H+1(2)_234TriOHPentDioic-2(2)				
2	---	1	C(10)H(14)Hf(1)O(14)	536.705
Hf+4_H+1(2)_Citric-3				
3	---	1	C(6)H(7)Hf(1)O(7)	369.609
Hf+4_H+1(2)_Lactic-1(2)				
4	---	1	C(6)H(12)Hf(1)O(6)	358.650
Hf+4_H+1(2)_Oxalic-2(2)				
2	---	1	C(4)H(2)Hf(1)O(8)	356.547
Hf+4_H+1(2)_TriMDTA-4(2)				
-2	---	1	C(22)H(30)Hf(1)N(4)O(16)	784.989
Hf+4_H+1(2)_TTHA-6				
0	---	1	C(18)H(26)Hf(1)N(4)O(12)	668.916
Hf+4_H+1(3)_Oxalic-2(3)				
1	---	1	C(6)H(3)Hf(1)O(12)	445.575
Hf+4_H+1(3)_PO4-3(3)				
-2	---	1	H(3)Hf(1)O(12)P(3)	466.431
Hf+4_H+1(4)_Oxalic-2(4)				
0	---	1	C(8)H(4)Hf(1)O(16)	534.602
Hf+4_H2O2				
4	---	1	H(2)Hf(1)O(2)	212.507

Hf+4 _HIMDA-2				
2	---	2	C(6)H(9)Hf(1)N(1)O(5)	353.633
Hf+4 _HIMDA-2(2)				
0	---	1	C(12)H(18)Hf(1)N(2)O(10)	528.774
Hf+4 _HOEDTA-3				
1	---	1	C(10)H(15)Hf(1)N(2)O(7)	453.730
Hf+4 _I-1				
3	---	1	Hf(1)I(1)	305.392
Hf+4 _I-1(4)_(g) Hafnium tetraiodide gas				
0	---	1	Hf(1)I(4)	686.092
Hf+4 _I-1(4)_(s) Hafnium tetraiodide				
0	---	2	Hf(1)I(4)	686.092
Hf+4 _IDA-2				
2	---	2	C(4)H(5)Hf(1)N(1)O(4)	309.580
Hf+4 _Imidazole				
4	---	1	C(3)H(4)Hf(1)N(2)	246.570
Hf+4 _Kojic-1				
3	---	1	C(6)H(5)Hf(1)O(4)	319.595
Hf+4 _Kojic-1(2)				
2	---	1	C(12)H(10)Hf(1)O(8)	460.699
Hf+4 _Lactic-1				
3	---	2	C(3)H(5)Hf(1)O(3)	267.563
Hf+4 _Lactic-1(2)				
2	---	1	C(6)H(10)Hf(1)O(6)	356.634

Hf+4_Maltol-1				
3	---	1	C(6)H(5)Hf(1)O(3)	303.596
Hf+4_Maltol-1(2)				
2	---	1	C(12)H(10)Hf(1)O(6)	428.700
Hf+4_Mucic-2				
2	---	1	C(6)H(8)Hf(1)O(8)	386.617
Hf+4_NO3-1				
3	---	2	Hf(1)N(1)O(3)	240.497
Hf+4_NO3-1_SO4-2				
1	---	1	Hf(1)N(1)O(7)S(1)	336.555
Hf+4_NO3-1_SO4-2(2)				
-1	---	1	Hf(1)N(1)O(11)S(2)	432.612
Hf+4_NO3-1(2)				
2	---	3	Hf(1)N(2)O(6)	302.502
Hf+4_NO3-1(3)				
1	---	3	Hf(1)N(3)O(9)	364.507
Hf+4_NO3-1(4)				
0	---	2	Hf(1)N(4)O(12)	426.512
Hf+4_NO3-1(5)				
-1	---	1	Hf(1)N(5)O(15)	488.517
Hf+4_NO3-1(6)				
-2	---	1	Hf(1)N(6)O(18)	550.521
Hf+4_NTA-3				
1	---	2	C(6)H(6)Hf(1)N(1)O(6)	366.609

Hf+4_OH-1				
3	---	5	H(1)Hf(1)O(1)	195.499
Hf+4_OH-1_EDTA-4				
-1	---	1	C(10)H(13)Hf(1)N(2)O(9)	483.713
Hf+4_OH-1_HIMDA-2				
1	---	2	C(6)H(10)Hf(1)N(1)O(6)	370.641
Hf+4_OH-1_IDA-2				
1	---	1	C(4)H(6)Hf(1)N(1)O(5)	326.587
Hf+4_OH-1(2)				
2	---	4	H(2)Hf(1)O(2)	212.507
Hf+4_OH-1(2)_HIMDA-2				
0	---	1	C(6)H(11)Hf(1)N(1)O(7)	387.648
Hf+4_OH-1(2)_PhenAcet-1(2)				
0	---	1	C(16)H(16)Hf(1)O(6)	482.791
Hf+4_OH-1(3)				
1	---	9	H(3)Hf(1)O(3)	229.514
Hf+4_OH-1(3)_234TriOHPentDioic-2				
-1	---	1	C(5)H(9)Hf(1)O(10)	407.613
Hf+4_OH-1(3)_Tartaric-2				
-1	---	1	C(4)H(7)Hf(1)O(9)	377.586
Hf+4_OH-1(4)				
0	---	3	H(4)Hf(1)O(4)	246.521
Hf+4_OH-1(5)				
-1	---	3	H(5)Hf(1)O(5)	263.529

Hf+4_OH-1 (6)				
-2	---	1	H (6) Hf (1) O (6)	280.536
Hf+4_Oxalic-2				
2	---	4	C (2) Hf (1) O (4)	266.512
Hf+4_Oxalic-2 (2)				
0	---	3	C (4) Hf (1) O (8)	354.531
Hf+4_Salicylaldoxime-2				
2	---	1	C (7) H (5) Hf (1) N (1) O (2)	313.614
Hf+4_SO4-2				
2	---	3	Hf (1) O (4) S (1)	274.550
Hf+4_SO4-2 (2)				
0	---	3	Hf (1) O (8) S (2)	370.607
Hf+4_SO4-2 (3)				
-2	---	1	Hf (1) O (12) S (3)	466.665
Hf+4_Tartaric-2				
2	---	2	C (4) H (4) Hf (1) O (6)	326.564
Hf+4_Tartaric-2 (2)				
0	---	1	C (8) H (8) Hf (1) O (12)	474.636
Hf+4_Tiron-4				
0	---	2	C (6) H (2) Hf (1) O (8) S (2)	444.689
Hf+4_Tiron-4 (2)				
-4	---	1	C (12) H (4) Hf (1) O (16) S (4)	710.886
Hf+4_Tiron-4 (3)				
-8	---	1	C (18) H (6) Hf (1) O (24) S (6)	977.083

Hf+4 _TriMDTA-4				
0	---	1	C (11) H (14) Hf (1) N (2) O (8)	480.733
Hf+4 (2) _OH-1 (2) _EDTA-4 (2)				
-2	---	1	C (20) H (26) Hf (2) N (4) O (18)	967.427
Hf+4 (2) _OH-1 (2) _HOEDTA-3 (2)				
0	---	1	C (20) H (32) Hf (2) N (4) O (16)	941.475
Hf+4 (4) _OH-1 (8)				
8	---	2	H (8) Hf (4) O (8)	850.027
Hf (s) Hafnium				
0	7440-58-6	22	Hf (1)	178.492
HfB2 (s) Hafnium diboride				
0	---	1	B (2) Hf (1)	200.112
HfC (s) Hafnium carbide				
0	---	1	C (1) Hf (1)	190.503
HfN (s) Hafnium nitride				
0	---	1	Hf (1) N (1)	192.499
HfO+2				
2	---	1	Hf (1) O (1)	194.491
HfO (OH) 2 (s) Hafnium oxide hydrate ?				
0	---	1	H (2) Hf (1) O (3)	228.506
HfO2 (am. , s)				
0	---	2	Hf (1) O (2)	210.491

HfO2 (s) Hafnium oxide; Hafnium dioxide; Hafnium dioxide, white cubic				
0	12055-23-1	5	Hf(1)O(2)	210.491
HfSrO3 (s) Hafnium strontium trioxide				
0	---	1	Hf(1)O(3)Sr(1)	314.110
HIMDA-2				
-2	---	201	C(6)H(9)N(1)O(5)	175.141
I-1 Iodide ion				
-1	20461-54-5	894	I(1)	126.900
I2 (s) Iodine; Iodine, orthorhombic				
0	7553-56-2	244	I(2)	253.800
IDA-2 Iminodiacetate ion				
-2	28528-43-0	428	C(4)H(5)N(1)O(4)	131.088
Imidazole Imidazole; 1,3-Diazole; 1H-Imidazole; im; Him				
0	288-32-4	422	C(3)H(4)N(2)	68.0782
Kojic-1				
-1	---	100	C(6)H(5)O(4)	141.103
Lactic-1 Lactate ion				
-1	---	643	C(3)H(5)O(3)	89.0709
Maltol-1 Maltolate ion				
-1	---	59	C(6)H(5)O(3)	125.104
N2 (g) Nitrogen gas; Nitrogen				

0	7727-37-9	566	N(2)	28.0134
NO3-1 Nitrate ion				
-1	14797-55-8	573	N(1)O(3)	62.0049
NTA-3 Nitritotriacetate ion; N,N-bis(carboxymethyl)glycinate ion; Triglycollamate ion; Triglycinate ion; NTA ion				
-3	28528-44-1	751	C(6)H(6)N(1)O(6)	188.117
O2(g) Oxygen gas; Oxygen				
0	7782-44-7	2559	O(2)	31.9988
OH-1 Hydroxide ion				
-1	14280-30-9	6558	H(1)O(1)	17.0073
Oxalic-2 Oxalate ion				
-2	338-70-5	1241	C(2)O(4)	88.0196
PhenAcet-1 Phenylacetate ion; 2-Phenylacetate				
-1	---	36	C(8)H(7)O(2)	135.142
Salicylaldoxime-2				
-2	---	35	C(7)H(5)N(1)O(2)	135.122
SO4-2 Sulfate ion; Sulphate ion				
-2	14808-79-8	1270	O(4)S(1)	96.0576
Sr(s) Strontium; Strontium, cubic; Strontium, alpha				
0	7440-24-6	55	Sr(1)	87.6200
Tartaric-2 Tartrate ion				
-2	---	393	C(4)H(4)O(6)	148.072

Tiron-4

4,5-Dihydroxy-1,3-benzenedisulphonate ion; 4,5-Dihydroxy-1,3-benzenedisulfonate ion;
 1,2-Dihydroxybenzene-3,5-disulphonate ion; 4,5-Dihydroxybenzene-1,3-disulphonate ion;
 Disodium 3,5-pyrocatecholdisulfonate ion; PDS

-4	77310-82-8	263	C(6)H(2)O(8)S(2)	266.197
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TriMDTA-4

-4	---	107	C(11)H(14)N(2)O(8)	302.241
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